Advanced Signal Processing and Machine Learning Methodologies for Contactless Vital Sign Extraction via Bed-Mounted Geophone Sensors

Introduction to Contactless Ballistocardiography and the BedDot Ecosystem

The continuous, long-term monitoring of human vital signs—specifically heart rate (HR), respiratory rate (RR), and blood pressure (BP)—has historically been confined to clinical environments utilizing obtrusive, contact-based modalities. Electrocardiography (ECG), photoplethysmography (PPG), and sphygmomanometry, while highly accurate, present profound limitations for continuous deployment in at-home geriatric care, sleep studies, and continuous ward monitoring. These methods disrupt circadian rhythms, induce patient discomfort, and suffer from data loss due to sensor detachment. In response to these limitations, the paradigm of physiological measurement has shifted aggressively toward engagement-free, contactless monitoring systems.[1] Among the most mathematically and physically sophisticated of these technologies is the bed-mounted geophone sensor, a highly sensitive seismometer capable of detecting and isolating physiological micro-vibrations generated by the human body and propagated through a bed frame.[1]

The theoretical underpinning of this technology relies on the physiological principles of ballistocardiography (BCG) and seismocardiography (SCG). According to Newtonian mechanics, the rapid acceleration of blood mass ejected from the left ventricle into the ascending and descending aorta during cardiac systole induces a reciprocal, low-amplitude mechanical displacement of the entire human body.[4] Simultaneously, the biomechanical expansion and contraction of the thoracic cavity during the respiratory cycle generates slow, low-frequency physical oscillations.[1] When a subject is recumbent on a mattress, these minute mechanical forces propagate through the viscoelastic medium of the mattress and the rigid structure of the bed frame, manifesting as complex, overlapping seismic waves collectively termed a bedseismogram (BSG).[3]

Systems engineered to capture this phenomenon, such as the BedDot geophone sensor (Model: Genesis 1.0), capitalize on ultra-sensitive seismic data acquisition to continuously compute HR, RR, and BP without requiring physical contact.[1] The BedDot device, measuring 92 mm x 63.5 mm x 53.48 mm and weighing 300 grams, is magnetically mounted beneath the bed frame, making it entirely invisible and unobtrusive to the patient.[1] Despite its physical simplicity, the extraction of clinical-grade physiological metrics from a raw BSG signal represents a

profoundly complex signal processing challenge. The raw waveform is inherently corrupted by environmental noise, macro-movements of the subject, and, frequently, the overlapping seismic vibrations of multiple bed occupants.[1] Furthermore, translating raw mechanical displacement into precise, continuous hemodynamic metrics like blood pressure necessitates the integration of advanced artificial intelligence and deep causal time-series learning architectures.[3]

The application of this technology is specifically indicated for adults over 12 years of age in both hospital and home settings, providing accuracy equivalent to the standard auscultation method and meeting the stringent error requirements of the ISO 81060-2:2018 standard.[1] However, the physical limitations of BSG signal acquisition dictate specific contraindications. The technology is generally unsuitable for patients with severe arrhythmias (such as atrial fibrillation or ventricular tachycardia), as highly irregular cardiac intervals disrupt the periodicity assumptions foundational to the signal processing algorithms.[1] Extreme obesity or being severely underweight can similarly alter the damping coefficient of the mattress, distorting the seismic wave propagation. Patients with sleep apnea syndrome may present prolonged respiratory pauses that complicate RR estimation, while environments with persistent, heavy mechanical vibrations can overwhelm the geophone's noise floor.[1]

The following analysis provides an exhaustive examination of the end-to-end mathematical, physical, and computational pipeline required to extract vital signs from raw geophone sensor data. This includes low-level binary payload parsing, multidimensional signal quality assessment, feature extraction, blind source separation, and deep learning architectures for hemodynamics.

Hardware Architecture and Sensor Physics

The foundation of the contactless vital sign monitoring pipeline is the hardware interface responsible for capturing physical vibrations and digitizing them for computational analysis. The primary sensor utilized in the BedDot system is the commercial off-the-shelf moving-coil geophone, specifically variants akin to the SM-24 geophone element.[1] A geophone consists of a magnetic mass suspended by delicate springs within a wire coil. As the bed frame vibrates in response to cardiac or respiratory ballistic forces, the magnetic mass remains relatively stationary due to its inertia, while the coil moves in tandem with the bed frame. This relative kinetic motion induces a proportional electrical voltage according to Faraday's Law of Induction.[2]

These geophones exhibit a natural resonant frequency of approximately 10 Hz, meaning they naturally function as second-order high-pass filters.[2] They are highly sensitive to the sudden, transient ballistic forces of heartbeats (which possess higher-frequency harmonic components) but are naturally insensitive to the slow, low-frequency oscillations of human respiration, which typically occur between 0.2 Hz and 0.4 Hz.[1] This physical reality necessitates highly specialized mathematical workarounds, such as amplitude demodulation, to extract

respiratory data.

The orientation of the geophone is a critical factor in signal integrity. While it might intuitively seem optimal to deploy horizontal geophones to capture the head-to-foot ballistic force of aortic blood flow, empirical physics dictates otherwise. A standard bed frame acts as an underdamped oscillator in the horizontal ($x - y$) plane. A single horizontal mechanical impulse can cause oscillations that reverberate for more than 1.8 seconds.[1] Because a standard human cardiac cycle lasts approximately 0.8 to 1.0 seconds, horizontal reverberations from a preceding heartbeat will catastrophically overlap with the subsequent heartbeat, destroying the signal morphology. Conversely, the bed is heavily damped in the vertical (z) direction against the floor. A vertical geophone impulse decays within approximately 0.3 seconds, preserving the distinct morphology of individual cardiac cycles.[1] Therefore, vertical geophones are exclusively utilized in advanced BSG systems.[2]

To process these micro-voltages, the analog signal is passed through an amplification circuit—typically with an amplification factor of 200—and fed into a high-precision 12-bit Analog-to-Digital Converter (ADC).[1] The ADC is typically interfaced with a single-board computer, such as a Raspberry Pi 3B or an equivalent onboard processor, which handles the localized digital signal processing and subsequent transmission via standard Wi-Fi protocols (such as WPA2 encrypted 802.11 b/g/n).[1] To capture the precise temporal dynamics required for both basic vital sign extraction and advanced blood pressure morphology analysis, the ADC samples the amplified geophone voltage at a continuous frequency of 100 Hz.[3] According to the Nyquist-Shannon sampling theorem, a 100 Hz sampling rate provides a usable frequency bandwidth of up to 50 Hz, which is vastly more than sufficient to capture the sub-10 Hz fundamental frequencies of human vital signs while retaining the high-frequency harmonics necessary for deep learning feature extraction.[1]

Raw Binary Data Acquisition and Low-Level Parsing

The digitized data output by the onboard ADC is rarely transmitted or stored in human-readable ASCII formats like CSV or JSON. For high-throughput, continuously operating sensor systems, data is encoded as packed binary streams to minimize memory footprint, preserve bandwidth, and drastically reduce computational overhead during transmission.[9]

The binary payload generated by the geophone arrays is typically structured as a continuous series of 32-bit signed integers. In a standard multi-channel system, these integers are arranged sequentially to represent a temporal index followed by the sensor readings, formatted strictly as: Time, Channel 1, Channel 2, and Channel 3.[9]

Understanding the system's endianness and binary representation is paramount for accurate data reconstruction. The architecture utilizes a little-endian byte ordering convention.[10] In a

little-endian memory architecture, the least significant byte (LSB) of a multi-byte numerical value is stored at the lowest memory address, while the most significant byte (MSB) is stored at the highest memory address. Furthermore, the 32-bit integers are signed, utilizing two's complement notation.[10] This allows the system to accurately represent both positive and negative voltage fluctuations, which directly correspond to the bidirectional physical displacement of the geophone's internal coil.[13]

To extract and reconstruct this data algorithmically, software environments such as Python require specific decoding protocols. The raw byte stream cannot be interpreted directly; it must be explicitly unpacked into computational primitives. When processing this data via sequential or stream-based software layers, the standard Python struct library is heavily utilized. The format string <i is applied to the unpacking function.[14] The < character explicitly designates the little-endian byte order, overriding the system's native byte order, while the i character dictates the extraction of a standard 4-byte (32-bit) signed integer.[15] The execution of struct.unpack('<i', byte_chunk) translates the raw binary payload into a computable integer format.[14]

However, for batch processing or historical data analysis—such as analyzing an entire 8-hour sleep session—processing the data byte-by-byte using struct introduces severe computational bottlenecks.[18] In these scenarios, vectorized operations provide vastly superior computational efficiency. The Python numpy library facilitates this exact requirement through the numpy.fromfile() function.[19] By specifying the data type parameter as dtype='<i4' (or equivalently np.int32 executed on a little-endian host architecture), the software can instantly map the entire binary file from the disk into a highly optimized, multidimensional numerical array in C-contiguous memory.[12] This vectorization is not merely an optimization; it is an absolute necessity when processing the millions of data points generated by continuous 100 Hz sampling over multi-hour sleep monitoring sessions.

Signal Quality Assessment and Bed Occupancy Detection

Once the raw 32-bit integer data is unpacked into a time-series vector, the signal must undergo rigorous quality assessment and preprocessing. The raw bedseismogram is heavily corrupted by thermal noise from the analog circuitry, high-frequency environmental vibrations (such as HVAC systems, washer/dryers, or passing vehicles), and extreme low-frequency baseline wander caused by minute shifts in the subject's center of mass.[2]

Bed Occupancy Determination via Multiple Feature Fusion

Determining whether the bed is actually occupied is the fundamental prerequisite for all subsequent calculations. Advanced systems employ a Multiple Feature Fusion (MFF) methodology to accurately detect bed entry and exit, circumventing the unreliability of simple

amplitude-based triggers or pressure pads.[1] The MFF approach mathematically combines three distinct signal metrics to achieve an occupancy detection accuracy exceeding 99% [2]:

1. **Spectral Entropy (SE):** This metric evaluates the complexity and energy distribution of the signal in the frequency domain. For a discrete signal $x(n)$, the discrete Fourier transform provides the power spectrum $S(m)$. This spectrum is normalized to represent a probability distribution $P(m)$. The Shannon entropy is then calculated as $SE = -\sum P(m) \log_2 P(m)$.[1] An empty bed subject only to ambient room noise presents a highly random, high-entropy frequency spectrum. Conversely, an occupied bed contains the highly periodic, structured rhythms of the human cardiac and respiratory systems, which significantly lowers the spectral entropy.[1]
2. **Kurtosis:** Kurtosis is the fourth standardized moment of a probability distribution, mathematically defined as the expected value of the standardized data raised to the fourth power.[1] Kurtosis detects the sudden, heavily tailed impulse forces associated with the physical act of sitting on, dropping onto, or exiting the bed.[1]
3. **Teager Energy Operator (TEO):** The TEO tracks the instantaneous energy of the oscillating system. Derived from a second-order differential equation, the total energy of oscillation is calculated as the sum of kinetic and potential energies, reacting rapidly to the sudden changes introduced by a human mass interacting with the mattress.[1]

By fusing these three features, the system reliably determines bed occupancy within a 30 to 60-second window, generating specific system alerts (such as internal HTTP API code 10 for "On Bed Alert" and code 20 for "Off Bed Alert").[1]

Movement and Posture Change Recognition

No vital sign extraction algorithm can succeed if the input signal is heavily distorted by large-scale body movements, such as a patient rolling over, coughing, or repositioning an arm. Body movements generate seismic waves with amplitudes multiple orders of magnitude larger than a heartbeat. A standard ballistic heartbeat typically generates a signal amplitude around 10^5 (in arbitrary uncalibrated units), whereas a gross body movement generates an amplitude exceeding 10^7.[1]

To preserve the integrity of the physiological data pipeline, these movement artifacts must be detected, and the corresponding temporal segments must be explicitly excluded from vital sign estimation.[2] A localized thresholding mechanism identifies these massive amplitude spikes. However, amplitude thresholding alone is insufficient, as it may misclassify localized structural noise. Therefore, the aforementioned Kurtosis metric is continuously monitored. In a normal, resting BSG signal, the distribution of amplitude values is relatively Gaussian. However, a sudden movement introduces extreme outliers into the signal window, creating a heavily

skewed distribution with thick tails. Empirical analysis establishes that a Kurtosis value greater than a threshold of 5 reliably indicates the presence of a macro-movement.[2] When the Kurtosis exceeds this threshold, the algorithm flags the segment, temporarily suspends HR and BP estimations, and logs the event for sleep quality analysis.[1]

Beyond mere movement detection, advanced BSG systems evaluate full posture changes. When a movement is detected, the algorithm analyzes a 5-second window of the signal immediately preceding the movement and a 5-second window immediately following it. The system computes a similarity average based on four metrics: Spectral Entropy, Kurtosis, TEO, and Power Spectral Density (PSD).[1] If the similarity index falls below an empirically derived threshold of 0.85, the system registers a definitive posture change.[1] The system is capable of distinguishing between four primary postures—laying on the back, chest, left side, and right side—with an average accuracy of 92.08%.[1] This capability is critical because the morphology of the BCG waveform alters significantly based on the subject's posture, and downstream blood pressure algorithms must account for these positional shifts.[22]

Fall From Bed Detection

A unique and highly critical application of the geophone system is the automated detection of falls from the bed, a major risk factor in senior care facilities.[1] The system continuously monitors the sequence of events. If a massive movement is detected, followed immediately by a transition to the "Off Bed" state, the system triggers a fall-down candidate classification routine.[1]

To distinguish a true fall from a rapid, intentional exit from the bed, the system extracts high-dimensional features from the localized signal event. In the time domain, these features include absolute signal amplitude, total event duration, the number of discrete sub-events, and the temporal distribution of the final 20 signal peaks.[1] In the frequency domain, the Power Spectral Density (PSD) is calculated via the Fast Fourier Transform to analyze the exact distribution of impact energy.[1]

These normalized features are fed into a supervised machine learning classifier. While K-Nearest Neighbors (KNN) and Gaussian Process (GP) classifiers exhibit acceptable accuracy, the Support Vector Machine (SVM) utilizing a non-linear kernel is overwhelmingly preferred for edge deployments due to its optimal balance of high accuracy (90.92% on raw data, reaching 97% in simulated real-world environments) and exceptionally low classification run-time (0.472 seconds on a Raspberry Pi), ensuring near-instantaneous alerts.[1]

Advanced Denoising and Signal Stabilization Pipelines

For the periods where the subject is confirmed to be on the bed and entirely stationary, the signal is advanced to the physiological filtering stage. The raw voltage equivalents undergo a z-score transformation, standardizing the signal to a mean of zero and a standard deviation of

one.[2] This normalization mitigates discrepancies caused by different patient weights and mattress compositions.

To eliminate ambient machinery noise (such as air conditioners operating at specific rotational frequencies), the system performs spectrum scanning and applies narrow notch filters to suppress iso-dominant frequencies.[1] For the extraction of vital signs, the frequencies of interest lie within a highly specific band: 0.1 Hz to 8 Hz.[1] Traditional systems rely on high-order Butterworth bandpass filters to isolate this range.[1] However, standard linear Infinite Impulse Response (IIR) filters introduce phase delay, temporally shifting the complex waveforms of the cardiac cycle. To counteract this, systems deploy forward-backward digital filtering (often implemented as the filtfilt function in Python). This technique passes the signal through the filter coefficients in the forward direction, reverses the resulting sequence, and passes it through the filter again, perfectly canceling out all phase distortion and squaring the amplitude response.[1]

A more mathematically sophisticated and increasingly preferred approach for BSG denoising is the Discrete Wavelet Transform (DWT).[2] Unlike Fourier transforms, which localize signals strictly in the frequency domain and lose temporal resolution, wavelets provide optimal, multi-resolution analysis in both the time and frequency domains. The choice of the mother wavelet is critical. Empirical studies establish that the Daubechies 12 (db12) and Symlet (sym10) wavelets provide the highest efficacy for BCG and ECG denoising.[2] The fractal, inherently asymmetric morphology of these wavelets closely mirrors the sharp, transient physical impulse of the cardiac ejection phase.

During DWT, the time-series signal is passed through iterative cascaded layers of low-pass and high-pass filters. This process decomposes the data into approximation coefficients (representing low-frequency, macro-trends) and detail coefficients (representing high-frequency, transient micro-events).[26] Baseline wander—the slow, low-frequency drift of the signal baseline caused by minute shifts in posture or the gradual settling of the mattress springs—is captured entirely within the highest-level approximation coefficients. By selectively setting these specific approximation coefficients to absolute zero and performing an Inverse Discrete Wavelet Transform (IDWT), the system reconstructs a perfectly stabilized, zero-mean signal that retains all the sharp, high-frequency morphological peaks of the cardiac cycle while completely eliminating baseline drift.[26]

Methodologies for Heart Rate (HR) Extraction

With the signal denoised and stabilized, the system initiates heart rate extraction. A standard resting heart rate falls between 40 and 100 beats per minute (approximately 0.66 Hz to 1.66 Hz).[1] The system estimates the HR every 3 seconds utilizing a rolling 15-second time window.[1] While calculating the periodicity of a repeating signal seems mathematically trivial, the BCG waveform generated by the geophone is notoriously complex and irregular.

Envelope-Based HR Estimation

Unlike an electrocardiogram (ECG), which presents a sharp, electrically driven, unmistakable R-peak, a seismocardiogram or ballistocardiogram waveform is multi-modal and mechanically driven. A single cardiac cycle generates a sequence of physical waves—traditionally labeled the H, I, J, K, L, and M waves—representing the complex, overlapping mechanical reverberations of ventricular contraction, blood ejection into the aorta, and the subsequent vascular recoil.[4] The J-wave is typically the highest amplitude peak. However, depending entirely on the subject's sleeping posture and the acoustic impedance of the mattress, the relative amplitudes of these waves can invert. In some postures, the diastolic peaks may exceed the systolic peaks, causing standard local maxima peak-detection algorithms to routinely double-count a single heartbeat.[2] Furthermore, the temporal spacing between heartbeats is not strictly periodic due to the autonomic nervous system's regulation of natural Heart Rate Variability (HRV).[2]

To counter this morphological chaos, advanced signal processing abandons direct peak detection on the raw waveform. Instead, the algorithm relies on extracting the mathematical envelope of the heartbeat cycle.[2] By applying techniques such as the Hilbert transform, the system calculates the analytic signal, effectively isolating the instantaneous amplitude envelope. This process creates a smoothed, overarching continuous boundary over the jagged, intricate waveform of the individual sub-waves. It effectively merges the complex I, J, and K wave cluster of a single cardiac systole into one broad, distinct, unmistakable mathematical peak.[2]

Empirical Truncated Statistics

Once the smoothed envelope is generated, a local maxima search identifies the distinct peaks corresponding to individual heartbeats. However, due to lingering environmental noise or subtle physiological shifts, falsely picked peaks and missed peaks are inevitable. Falsely picked peaks artificially decrease the calculated inter-beat interval (IBI), falsely raising the reported HR. Conversely, missed peaks artificially increase the IBI, falsely lowering the reported HR.[29]

To achieve clinical-grade stability in the estimation, a novel empirical truncated statistics analysis is deployed.[1] Let the random variable X represent the interval duration between two sequentially picked peaks in the envelope. The algorithm defines a dynamic temporal window and mathematically removes extreme statistical outliers (specifically, the lowest 10% and highest 10% of the recorded intervals) before calculating the true average inter-beat interval. Mathematically, the expected value of the heartbeat period is defined as a truncated average:

$$E(X|F^{-1}(a) < X \leq F^{-1}(b)) = \frac{\int_a^b xg(x)dx}{F(b) - F(a)}$$

Where F^{-1} represents the inverse cumulative distribution function of the inter-beat intervals, a and b are the strictly defined lower and upper percentile bounds (typically defined as $a = 0.1$ and $b = 0.9$), and $g(x)$ is the probability density function of the intervals restricted exclusively to this truncated bound.[1]

This statistical truncation isolates the true, central tendency of the heart rate, completely immunizing the final calculation against the occasional false-positive or false-negative peak detection. Utilizing this methodology, systems demonstrate an exceptionally low mean absolute error (MAE) of 1.38 to 2.41 beats per minute when benchmarked against gold-standard FDA-approved wearable devices, operating effortlessly across highly variable heart rates ranging from 40 to 150 BPM.[1] If the calculated heart rate drops below or exceeds strictly defined clinical thresholds, the software generates highly specific alert codes (e.g., Code 30: Heart Rate Too High, Code 40: Heart Rate Too Low).[1]

Multi-Occupant Separation via Blind Source Separation

A profound and unique challenge in contactless bed-mounted geophone monitoring is the "shared bed" scenario, wherein two subjects occupy the same mattress. The vibration signals generated by both subjects propagate simultaneously through the bed frame and overlap perfectly in the time domain, creating an incredibly chaotic mixed signal.[1] Because both subjects possess heart rates that exist in identical, narrow frequency bands (e.g., Subject 1 at 62.8 BPM and Subject 2 at 59.2 BPM), standard frequency-domain bandpass filtering is entirely useless for separating them.[1]

This scenario represents a classic "cocktail party problem" applied to solid-state seismic propagation. To resolve this, systems like VitalMon utilize multiple geophones (typically one mounted under each side of the bed frame) and employ an advanced mathematical algorithm known as the Degenerate Unmixing Estimation Technique (DUET).[1] DUET separates overlapping signals by analyzing their unique spatial signatures—specifically, the symmetric attenuation (the loss of signal amplitude as it travels through the mattress) and the relative time delay (the phase shift) as the seismic waves travel from the physical source (the beating heart) to the spatially separated receivers.[1]

The algorithm partitions the 40-second data window and calculates the Short-Time Fourier Transform (STFT) for the signals captured by geophone 1 (x_1) and geophone 2 (x_2). In the resulting time-frequency domain, the relative attenuation and relative phase delay are computed mathematically for every single frequency bin.[1]

A critical, non-negotiable requirement for the DUET algorithm to succeed is W-disjoint

orthogonality; meaning the spectral energy from one source must completely dominate the other source at any given specific frequency and time coordinate. The mathematical condition is strictly defined as:

$$\hat{s}_1(\tau, \omega)\hat{s}_2(\tau, \omega) = 0, \forall \tau, \omega$$

Where $\hat{s}_1$ and $\hat{s}_2$ are the time-frequency representations of the two individual heartbeat sources.[1] Standard resting heartbeats, unfortunately, heavily violate this assumption. Their fundamental frequencies are physically too close together, and they critically lack short-term frequency stability (due to natural heart rate variability).[1]

To mathematically force W-disjoint orthogonality and achieve clean signal separation, the algorithm makes two critical procedural adjustments. First, instead of looking at the low fundamental frequency of the heartbeat (e.g., 1 Hz), the algorithm focuses its analysis on the high-frequency harmonics of the heartbeat (e.g., the 8th harmonic at 8 Hz and 8.8 Hz). At higher harmonics, the absolute frequency difference between the two subjects is linearly multiplied, providing a much larger mathematical distance for separation.[1] Second, the algorithm slices the temporal processing window into extremely short segments of exactly 0.7 seconds. Within a micro-window of 0.7 seconds, it is highly statistically improbable for the harmonic peaks of both subjects to overlap perfectly in time, ensuring the W-disjoint orthogonality condition is satisfactorily met.[1]

The system then plots a 3-dimensional energy histogram mapping the relative delay on the x-axis, the symmetric attenuation on the y-axis, and the localized spectral energy on the z-axis. A binary mask is applied in this 3D space to cluster the peaks based on spatial location. For instance, positive attenuation clusters mathematically belong to the occupant on the right side, while negative attenuation clusters belong to the occupant on the left. This binary masking cleanly extracts two distinct, pristine heartbeat vectors from the chaotic mixed waveform, allowing both individuals' heart rates to be calculated simultaneously with a median error of merely 0.72 BPM.[1]

Methodologies for Respiratory Rate (RR) Extraction

Extracting the respiratory rate from a moving-coil geophone presents a profound physical paradox: the geophone is physically designed by its own mass-spring mechanics to ignore respiration. As established, a geophone has a natural frequency of 10 Hz and acts as a steep second-order high-pass filter.[1] Human respiration, however, occurs at a very low frequency of 0.13 Hz to 0.66 Hz (8 to 40 breaths per minute).[1] Furthermore, the physical velocity of the thoracic cavity slowly expanding during inhalation is exceptionally low compared to the explosive, high-velocity ballistic impact of a heartbeat.[1] Consequently, the direct fundamental frequency of the respiration signal is virtually invisible in the raw geophone frequency

spectrum; it is buried entirely under the noise floor and the sensor's own attenuation curve.[1]

To extract the respiratory rate, the advanced signal processing algorithms do not look for the respiration signal directly; rather, they mathematically hunt for the *physical effect* of respiration on the high-frequency heartbeat signal.

Amplitude Modulation (AM) and Square-Law Demodulation

As a subject inhales and exhales, the physical volume of air within the thoracic cavity fluctuates drastically. This cyclical fluctuation in air volume continually alters the effective stiffness, density, and acoustic impedance of the human chest wall.[1] As the high-frequency ballistic forces of the heartbeat propagate outward through the chest tissue toward the mattress and bed frame, they encounter this constantly changing acoustic impedance.

Consequently, the physical amplitude of the heartbeat vibrations reaching the geophone is noticeably larger during exhalation (due to lower tissue impedance) and smaller during inhalation (due to higher tissue impedance). This physiological reality dictates that the high-frequency heartbeat signal acts perfectly as a carrier wave, and the low-frequency respiratory signal acts as the information wave, physically modulating the amplitude of the carrier. The raw geophone signal can therefore be perfectly mathematically modeled as an Amplitude Modulation (AM) communications system [1]:

$$s(t) = s_r(t) \sum_{j=1}^{N} s_{h_j}(t)$$

Where $s(t)$ is the observed raw geophone signal, $s_r(t)$ is the low-frequency respiration signal, and $s_{h_j}(t)$ represents the heartbeat fundamental frequency and its manifold higher-order harmonics.[1]

Extracting the exact respiratory rate requires accurately demodulating this AM signal. Traditional RF envelope detectors or product detectors fail entirely in this biological context because the carrier frequency (the human heart rate) is not constant; it fluctuates beat-to-beat due to HRV.[1] To mathematically bypass this limitation, advanced systems deploy a Square-Law Demodulation (SLD) algorithm.[1]

In the SLD process, the geophone signal is mathematically multiplied by itself (squared). Squaring an amplitude-modulated signal shifts the internal frequency components. According to trigonometric identities, squaring the combined carrier and modulating waves yields a new, unmodulated low-frequency component that oscillates at exactly twice the frequency of the

respiration signal ($2f_r$), stripped entirely of the high-frequency heartbeat carrier.[1]

Once the entire time-series signal is squared, a strict, high-order low-pass filter with a cutoff frequency of approximately 0.6 Hz is applied. This completely eliminates all the remaining high-frequency energy of the squared heartbeats and harmonics, leaving only the isolated, low-frequency baseband respiration signal.[1] The square root of the signal is taken to restore magnitude scaling, and finally, an Autocorrelation Function (ACF) is applied to the isolated baseband signal. The ACF evaluates the signal against delayed versions of itself to calculate the true mathematical periodicity of the waveform. This approach yields an extraordinarily precise respiratory rate estimation, boasting a mean estimation error of a mere 0.38 breaths per minute for single occupants and remaining robust even in shared-bed configurations.[1] Abnormal respiratory rates falling outside the 8-40 RPM clinical range immediately trigger HTTP API alerts (Code 50: RR High, Code 60: RR Low).[1]

Alternative Extrapolation Modalities

An alternative but equally robust computational methodology for deriving the respiratory envelope avoids the frequency-domain complexities of SLD by operating entirely within the time domain.[1] In this approach, once the individual heartbeat peaks are successfully identified by the HR extraction algorithm, the system performs a linear or cubic spline interpolation specifically between the local maxima (the extreme peaks) of successive heartbeats.[1] By mathematically stringing these discrete peaks together, the algorithm draws an artificial, continuous contour line over the top of the entire signal sequence.

Because the peak amplitudes rise and fall in perfect synchrony with the respiratory cycle due to thoracic impedance changes, this extrapolated contour line forms a perfect, low-frequency respiratory envelope.[1] Standard peak-finding algorithms and frequency analysis are then executed on this secondary, artificially constructed envelope to literally count the physical inhalations and exhalations. This method provides highly accurate RR tracking even during periods of heavy cardiac variability, yielding an average error of only 0.52 to 0.89 breaths per minute.[1]

Blood Pressure Estimation via Deep Time-Series Learning Architectures

The final, and most technologically demanding, capability of the advanced geophone monitoring pipeline is the continuous, contactless estimation of Systolic Blood Pressure (SBP) and Diastolic Blood Pressure (DBP) without any form of cuff inflation.[3] Traditional blood pressure monitoring requires the occlusion of a peripheral artery via an inflatable brachial cuff (sphygmomanometer), rendering continuous, nocturnal sleep monitoring highly uncomfortable and disruptive to the subject's natural circadian rhythms.[3] Deriving exact millimetric mercury (mmHg) pressure values from a contactless seismic vibration sensor requires moving beyond

classical deterministic signal processing and entering the realm of advanced deep learning and computational hemodynamics.

The Hemodynamic Basis of BCG-derived Blood Pressure

The theoretical link between the BSG/BCG waveform and actual systemic blood pressure relies entirely on the cardiovascular physics of vascular compliance, arterial stiffness, and pulse wave velocity. Traditional cuffless BP estimation relies heavily on calculating the Pulse Transit Time (PTT) or Pulse Arrival Time (PAT)—the exact temporal duration it takes for a pressure wave to travel from the contracting heart to a peripheral capillary location.[4] This is conventionally measured using an ECG to mark the initial ventricular contraction and a PPG sensor on the finger to mark the pulse's arrival.[35] The velocity of this pulse wave is directly, physically proportional to blood pressure; higher pressure stiffens the arterial walls, causing the mechanical pulse wave to travel significantly faster through the vasculature.[35]

In a purely contactless geophone system deployed under a mattress, peripheral sensors like PPGs are entirely unavailable. Instead, the AI models must exploit the subtle temporal phase differences and minute morphological shape changes inherent within the single complex BCG waveform itself.[4] For instance, the exact time delay between the initial ballistic force of left ventricular ejection (represented by the I-wave of the BCG) and the subsequent physical reflection of the pulse wave bouncing back from the descending aorta (represented by the K-wave) acts as a highly localized, intrinsic proxy for Pulse Transit Time.[4] Furthermore, the absolute amplitude and the steepness of the slope of the J-wave carry critical hemodynamic information regarding pulse pressure variations, vascular resistance, and left ventricular contractility.[1]

Early computational approaches to this problem relied on automatic delineation algorithms to explicitly, mathematically identify these specific fiducial points within the waveform.[22] The algorithms would calculate the exact Instantaneous Phase Difference (IPD) between specific peaks and feed these engineered features into standard regression models or simple artificial neural networks (ANNs).[36] However, these rigid feature-engineering methods proved fragile. They were heavily dependent on perfect, noise-free signal clarity and were easily derailed by the complex, highly variable nature of changing sleeping postures, which inherently alter the shape of the BCG wave.[22]

Deep Learning Architectures: From CNN-LSTM to VTCN

To achieve genuine clinical-grade accuracy that complies with the most stringent international standards—including the Association for the Advancement of Medical Instrumentation (AAMI), the British and Irish Hypertension Society (BHS) protocol, and the FDA-cleared ISO 81060-2:2018 requirements—modern contactless systems bypass manual feature extraction entirely.[1] Instead, they feed the minimally processed, high-dimensional 100 Hz BSG time-series directly into end-to-end deep neural networks capable of autonomously learning abstract,

nonlinear hemodynamic representations.[3]

CNN-LSTM Hybrid Models

The initial major breakthrough in this specific domain was the deployment of hybrid architectures combining Convolutional Neural Networks (CNN) with bidirectional Long Short-Term Memory (LSTM) recurrent networks.[3] In this architecture, 10-second data windows (comprising 1,000 discrete, high-resolution floating-point samples) are passed into 1D-CNN layers. The convolutional filters slide iteratively across the time-series, acting as automated, self-learning feature extractors. They learn to recognize the abstract morphological signatures of vascular compliance, such as the exact sharpness of the systolic upstroke or the depth of the dicrotic notch equivalent hidden in the seismic wave.[6]

The resulting high-dimensional spatial feature maps generated by the CNN are then fed sequentially into the LSTM layers. LSTMs are specialized Recurrent Neural Networks (RNN) equipped with complex internal gating mechanisms (input, output, and forget gates) mathematically designed to capture long-range temporal dependencies.[3] In the context of estimating continuous blood pressure, the LSTM tracks the crucial beat-to-beat variability and the overarching physiological trends occurring over the 10-second window, contextualizing a single heartbeat's morphology against the rhythmic patterns of the preceding beats.

Vital Temporal Convolutional Networks (VTCN)

While CNN-LSTM hybrid networks offer high predictive accuracy, recurrent networks are notoriously computationally expensive, slow to train, and severely struggle with parallelization, making them less than optimal for low-latency, real-time edge processing on limited hardware like a Raspberry Pi. To overcome these computational limits, state-of-the-art frameworks introduced the Vital Temporal Convolutional Network (VTCN).[3]

The VTCN architecture abandons the concept of mathematical recurrence entirely, relying instead on a highly modified causal convolutional framework optimized specifically for vital sign time-series data.[3] In traditional image-based convolutions, the mathematical filter aggregates data from both the past and the future relative to the current time step. In real-time continuous physiological monitoring, peering into the "future" is physically impossible and mathematically breaks the strict predictive causality of the model.[3]

Causal convolutions solve this by mathematically ensuring that the output at time t is convolved exclusively from data elements at time t and strictly earlier in the sequence.[3] To achieve the massive receptive field required to analyze a full 10-second (1000-step) cardiovascular history without requiring thousands of computationally heavy network layers, the VTCN employs dilated convolutions. Dilation introduces specific mathematical gaps between the kernel elements, allowing the receptive field of the network to expand exponentially with network depth. This allows a single network to capture both the immediate,

high-frequency cardiac morphology (the individual beat) and the long-term, low-frequency respiratory and autonomic nervous system trends (the 10-second history) simultaneously.[3]

The highly modified structure of the VTCN introduces specific, vital architectural enhancements over classical TCNs. The activations in the l-th layer and j-th block receive the direct input from the preceding block. Within every single convolutional block, the VTCN integrates a Batch Normalization layer immediately followed by a Rectified Linear Unit (ReLU) activation function.[3] Batch normalization mathematically standardizes the inputs to each individual layer. This massively accelerates mathematical convergence during model training and acts as a powerful statistical regularizer to prevent the network from merely overfitting to a specific patient's unique bedseismogram signature.[3] The ReLU activation introduces the necessary nonlinear mathematical capability while simultaneously preventing the vanishing gradient problem that plagues deep time-series models.[3]

Finally, critical residual connections span across the convolutional blocks. These residual links mathematically add the raw input of a block directly to its processed output. This allows the network to learn subtle modifications to the identity mapping rather than attempting to learn the entire complex transformation from scratch, preserving the pristine, underlying high-frequency features of the raw BSG signal deep into the later layers of the network.[3] At the terminus of the network, the high-dimensional latent representations are mathematically flattened and passed through a specialized linear regression head, outputting dual continuous numerical values representing the precisely estimated Systolic and Diastolic blood pressures in mmHg.[3]

Performance and Standardization Benchmarking

The deployment of the VTCN architecture yields unprecedented results in the domain of contactless hemodynamics. By exploiting the deep mathematical latent features of the bedseismogram rather than relying on explicit, fragile peak-to-peak timing, the model demonstrates remarkable robustness to varying sleep postures, diverse patient body masses, and different acoustic mattress compositions (e.g., innerspring versus high-density memory foam).[3]

When benchmarked directly against gold-standard, FDA-cleared continuous wearable devices, the VTCN architecture meets the rigorous evaluation criteria set forth by the British and Irish Hypertension Society (BHS) and the Association for the Advancement of Medical Instrumentation (AAMI).[3]

The comparative predictive efficacy of the VTCN against other leading deep learning time-series models underscores its architectural superiority:

Deep Learning Architecture	Structural Paradigm	Suitability for Real-Time BSG Blood Pressure Estimation
mWDN (Multilevel Wavelet Network)	Explicit Frequency Decomposition	Moderate; struggles with non-stationary phase shifts across highly variable bed frames and mattresses. [38]
RNN-FCN	Recurrence + Convolution	High predictive accuracy, but computationally heavy; suffers from severe latency in embedded edge deployments. [38]
CNN-LSTM	Hybrid Spatio-Temporal	Excellent temporal dependency tracking; strong limit of agreement (94.76% SBP, 95.19% DBP fall within limits). [3]
VTCN	Dilated Causal Convolution	Superior; zero future-leakage, highly parallelizable architecture. Achieves BHS Grade A with SBP MAE of 4.80 mmHg. [3]

The VTCN algorithm specifically achieves a Mean Absolute Error (MAE) of just 4.80 mmHg for Systolic pressure (well within the clinical acceptable range of 90-160 mmHg) and 2.57 mmHg for Diastolic pressure (clinical range 50-110 mmHg).[1] Under the strict BHS protocol, 88.02% of all Diastolic predictions fall within an absolute difference of $\leq$ 5 mmHg, securing a highly coveted Grade A rating.[3] Furthermore, 94.46% of Systolic and 95.18% of Diastolic predictions fall strictly within the statistical limits of agreement.[3] This performance mathematically and clinically validates the capability of advanced micro-vibrational analysis to safely, accurately, and continuously monitor cardiovascular health without any physical intrusion or patient engagement.

Synthesis and Clinical Implications

The extraction of highly accurate heart rate, respiratory rate, and continuous blood pressure from a bed-mounted geophone sensor represents a profound convergence of mechanical physics, low-level binary data engineering, and advanced artificial intelligence. The

computational pipeline begins at the most granular hardware level: parsing raw 32-bit signed little-endian binary streams output by a 100 Hz analog-to-digital converter, translating microscopic physical voltage changes into continuous mathematical data arrays.[12]

Through the strategic mathematical application of Discrete Wavelet Transforms, specifically leveraging the Daubechies 12 and Symlet 10 wavelets, the system completely eliminates baseline wander while perfectly preserving the transient ballistic forces of cardiac ejection.[2] By utilizing higher-order statistical moments such as Kurtosis, alongside spectral Shannon entropy and the Teager Energy Operator, the system autonomously identifies bed occupancy within seconds and aggressively discards macro-movement artifacts that would otherwise catastrophically corrupt the physiological calculations.[1]

The methodologies deployed to extract specific vital signs highlight the mathematical ingenuity required to overcome the strict physical limitations of the seismic sensor. The deployment of empirical truncated statistics on the mathematical signal envelope perfectly circumvents the inherent unreliability of raw BCG peak detection, resulting in highly stable heart rate metrics boasting an error of less than 2.41 BPM.[2] Furthermore, by mathematically framing the multi-occupant shared bed scenario as a blind source separation problem, the application of the Degenerate Unmixing Estimation Technique (DUET) successfully isolates overlapping heartbeats using highly specific spatial attenuation and relative phase delay metrics across multiple localized sensors.[1]

Perhaps most remarkably, the system physically overcomes the geophone's natural insensitivity to low-frequency human respiration by mathematically modeling the high-frequency cardiac signal as an amplitude-modulated carrier wave. The application of Square-Law Demodulation successfully translates the shifting acoustic impedance of the inflating thoracic cavity into a highly accurate, isolated respiratory rate, stripping away the cardiac carrier to reveal the underlying respiratory rhythm with an accuracy of within 0.38 breaths per minute.[1]

Finally, the evolutionary transition from classical, rigid feature-engineering to deep causal convolutional architectures, specifically the Vital Temporal Convolutional Network (VTCN), unlocks the ultimate clinical capability of continuous, contactless blood pressure monitoring.[3] By autonomously learning the deeply hidden hemodynamic morphologies embedded within the seismocardiogram, the VTCN provides clinical-grade blood pressure estimations that rival, and in terms of continuous patient comfort, exceed traditional occlusive cuffs.[3] Collectively, these advanced signal processing and machine learning methodologies establish the seismocardiographic geophone not merely as a novelty, but as a transformative, clinically viable instrument in the future of non-invasive, continuous health and sleep monitoring.

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